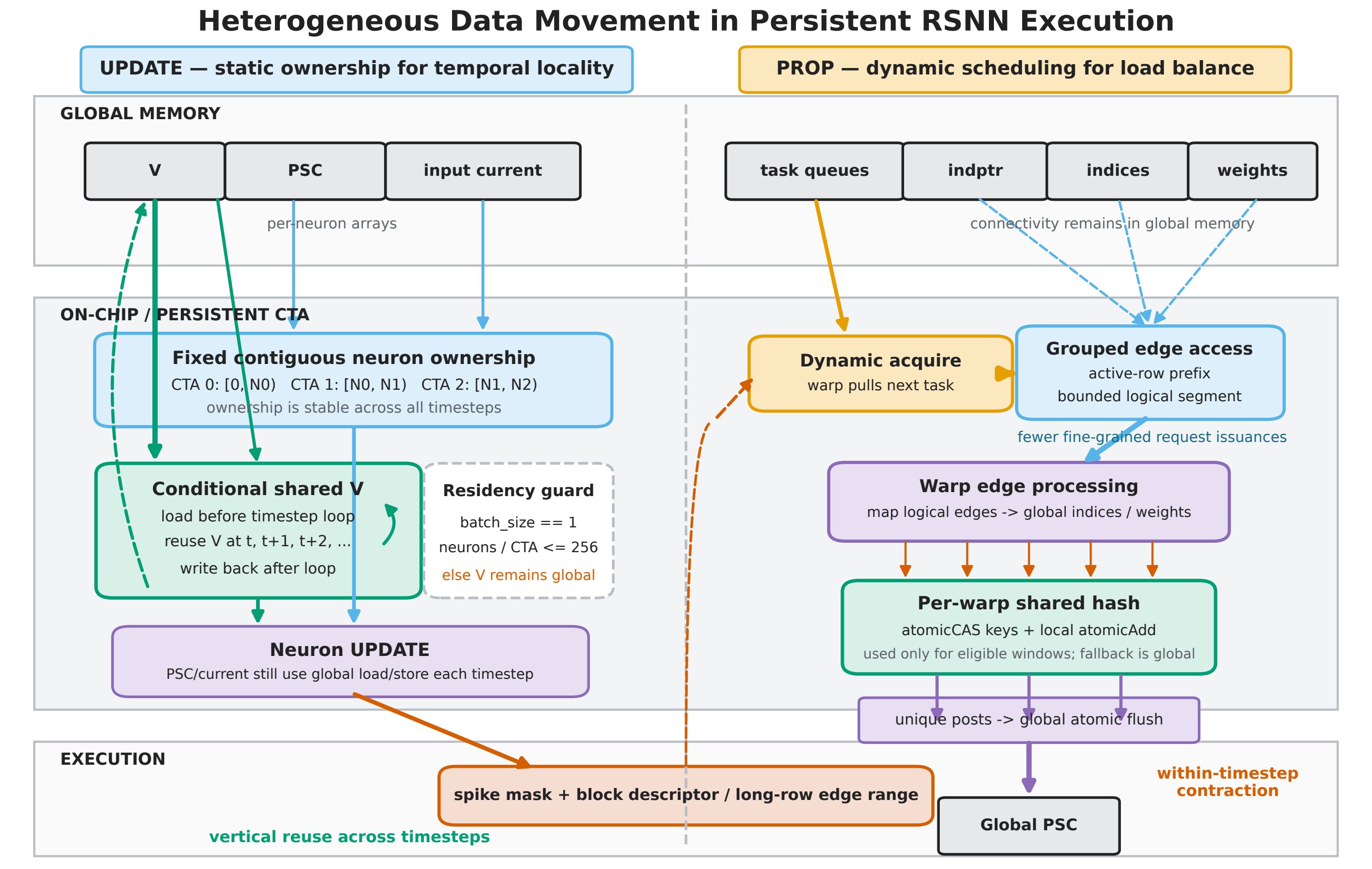
Heterogeneous Data Movement in Persistent RSNN Execution
UPDATE — static ownership for temporal locality
PROP — dynamic scheduling for load balance
GLOBAL MEMORY
V
PSC
input current
per-neuron arrays
task queues
indptr
indices
weights
connectivity remains in global memory
ON-CHIP / PERSISTENT CTA
Fixed contiguous neuron ownership
CTA 0: [0, N0) CTA 1: [N0, N1) CTA 2: [N1, N2)
ownership is stable across all timesteps
Conditional shared V
load before timestep loop
reuse V at t, t+1, t+2, ...
write back after loop
Residency guard
batch_size == 1
neurons / CTA <= 256
else V remains global
Neuron UPDATE
PSC/current still use global load/store each timestep
Dynamic acquire
warp pulls next task
Grouped edge access
active-row prefix
bounded logical segment
fewer fine-grained request issuances
Warp edge processing
map logical edges -> global indices / weights
Per-warp shared hash
atomicCAS keys + local atomicAdd
used only for eligible windows; fallback is global
unique posts -> global atomic flush
EXECUTION
spike mask + block descriptor / long-row edge range
within-timestep contraction
Global PSC
vertical reuse across timesteps